Notes: Hadlock and Pierce (RFS, 2010)

New Evidence on Measuring Financial Constraints: Moving Beyond the KZ Index

Contents

1 Big picture

- **Research question.** How should researchers measure financial constraints? In particular, does the popular Kaplan-Zingales (KZ) index actually identify financially constrained firms?
- **Main idea.** The paper builds a hand-collected benchmark measure of financial constraints from qualitative disclosures in firms' SEC filings. It then asks which commonly used quantitative variables line up with that benchmark.
- **Empirical object.** Hadlock and Pierce randomly sample Compustat firms from 1995 to 2004, read annual reports and 10-K filings, code statements about liquidity and financing ability, and assign each firm-year to a constraint category from not financially constrained to financially constrained.
- **Main finding.** The KZ index performs poorly. Only cash flow and leverage consistently enter with the same sign as the KZ index. Cash holdings, a key KZ component, usually enter with the opposite sign: firms with more cash look more constrained, consistent with precautionary saving.
- **Alternative index.** The paper proposes the size-age index, or SA index, based only on firm size and firm age. Small and young firms are more likely to be constrained; the relation becomes flat for very large or mature firms.
- **Why size and age?** Size and age are intuitive, robust predictors and are less endogenous than leverage, cash flow, dividends, payout, bond ratings, or cash holdings.
- **Validation.** Sorting firms by the SA index produces the Almeida, Campello, and Weisbach pattern: constrained firms have a significant cash-flow sensitivity of cash, while unconstrained firms do not.
- **Important warning.** Investment-cash flow sensitivities do not rise monotonically with constraint status. This confirms the Kaplan-Zingales critique of using investment-cash flow sensitivity as a constraint measure.

Economic takeaway. The paper argues that researchers should be very cautious with the KZ index and with measures based on endogenous financial choices. A simple size-age measure often has better empirical support and cleaner interpretation.

2 Incremental contribution of the paper

- **New hand-collected benchmark.** The paper follows the spirit of Kaplan and Zingales by reading management disclosures, but applies the approach to a larger, modern, randomly selected public-firm sample.
- **Avoids a hard-wiring problem.** The authors deliberately avoid using cash balances, dividends, and repurchases when assigning the qualitative constraint categories. This prevents the same quantitative variables from appearing mechanically in both the dependent variable and the regressors.
- **Direct evaluation of the KZ index.** The paper estimates ordered logit models parallel to the ones behind the KZ index and shows that the estimated coefficients differ sharply from the traditional KZ weights.

- **Explains why KZ can look good mechanically.** When the authors reintroduce quantitative information into the dependent variable, the estimates become much more supportive of KZ. This shows that the KZ index partly reflects a coding artifact.
- **Broad comparison of constraint proxies.** The paper evaluates KZ variables, Whited-Wu variables, Almeida-Campello-Weisbach sorting variables, bond-rating measures, commercial-paper-rating measures, and firm age.
- **New practical measure.** The paper proposes the SA index, a simple formula using only firm size and age. It is easy to compute in large samples and avoids many endogeneity concerns.
- **Corroboration from an independent method.** The SA index successfully separates firms that do and do not exhibit a cash-flow sensitivity of cash.
- **Clarifies what not to do.** The paper reinforces that investment-cash flow sensitivities and the KZ index should not be treated as reliable direct measures of financial constraints.

3 Data and measurement

3.1 Sample selection and filings

The starting point is the Compustat universe from 1995 to 2004. The authors exclude financial firms, regulated utilities, and firms incorporated outside the United States. They then sort by Compustat identifier and select every twenty-fourth firm, producing an initial random sample of 407 firms.

After requiring electronic filings, nonzero sales and assets, and enough data to compute KZ variables, the main sample contains 356 firms and 1,848 firm-years.

- **Disclosure sources.** The authors use annual reports and 10-K filings from Lexis-Nexis and SEC EDGAR.
- **Sections read.** They read the shareholder letter and management discussion and analysis.
- **Keyword search.** They also search filings for words such as financing, finance, investing, invest, capital, liquid, liquidity, note, covenant, amend, waive, violate, and credit.
- **Goal.** They extract statements about the firm's ability to raise funds, fund operations, or finance investments.

3.2 Statement-level constraint coding

Each extracted statement receives an integer code from 1 to 5. Higher values indicate more severe financial constraints.

- **Code 1.** The statement indicates excessive, strong, or more than sufficient liquidity.
- **Code 2.** The statement indicates adequate or sufficient liquidity.
- **Code 3.** The statement gives a soft warning, an opaque disclosure, or a possible future qualification, but no current financing problem.

- **Code 4.** The statement indicates a current liquidity problem, difficulty obtaining financing, or a postponed financing, but not a clear investment cut or severe distress.
- **Code 5.** The statement indicates clear financial trouble, such as a substantive covenant violation, a going-concern statement, loss of normal credit sources, renegotiated debt payments, or investment being affected by liquidity problems.

3.3 Firm-year constraint categories

The statement codes are aggregated into five mutually exclusive firm-year categories:

- **NFC:** not financially constrained.
- **LNFC:** likely not financially constrained.
- **PFC:** potentially financially constrained.
- **LFC:** likely financially constrained.
- **FC:** financially constrained.

Preferred classification. The paper's preferred categorization is qualitative scheme 2. It is the same as qualitative scheme 1 except that it ignores generic, boilerplate warnings about possible future liquidity problems. This matters because many firms issue vague legalistic warnings that do not appear informative about actual constraints.

3.4 Why the authors avoid quantitative coding

Kaplan and Zingales used both qualitative and quantitative information to classify firms. Hadlock and Pierce argue that this creates a mechanical problem when the classification is later regressed on the same quantitative variables.

For example, if high cash automatically moves a firm toward the unconstrained category, then a regression will mechanically find a negative coefficient on cash, even if qualitative disclosures say high-cash firms are more constrained because they save for precautionary reasons.

3.5 Key quantitative measures

The paper compares its qualitative constraint categories to standard quantitative variables:

- **KZ variables:** cash flow, Tobin's Q, leverage, dividends, and cash holdings, usually scaled by beginning-of-year property, plant, and equipment.
- **WW variables:** cash flow, a dividend-payer dummy, leverage, firm size, industry sales growth, and firm sales growth.
- **ACW sorting variables:** payout, firm size, no bond rating, and no commercial-paper rating.
- **Firm age:** the number of years since the firm first appears on Compustat with a non-missing stock price.

4 Methodology and empirical specifications

4.1 Ordered logit framework

The basic empirical model predicts a firm's qualitative constraint status using quantitative characteristics. The dependent variable is ordered from 1 for NFC to 5 for FC. A generic version of the ordered logit specification is:

$$Constraint^*_{it} = X'_{it}\beta + \varepsilon_{it}, \tag{1}$$

where observed categories are increasing functions of the latent constraint variable $Constraint^*_{it}$.

Interpretation. A positive coefficient means the variable is associated with more severe financial constraints. A negative coefficient means the variable is associated with fewer constraints.

4.2 Evaluating the KZ index

The traditional KZ index is:

$$KZ = -1.002\frac{CF}{K} + 0.283Q + 3.139\frac{Debt}{Capital} - 39.368\frac{Div}{K} - 1.315\frac{Cash}{K}. \tag{2}$$

The paper estimates ordered logit models using the five KZ components and compares the signs and significance of the estimated coefficients to the KZ weights.

- Cash flow and leverage usually match KZ: lower cash flow and higher leverage predict more constraints.
- Tobin's Q and dividends are mixed and often insignificant.
- Cash holdings usually enter positively, opposite of KZ. This means high-cash firms often look more constrained in qualitative disclosures.

Core diagnostic. An index created from the paper's baseline coefficients has approximately zero correlation with the traditional KZ index in the Compustat universe. This is a strong warning against KZ sorts.

4.3 The hard-wiring test

The authors also construct a qualitative/quantitative categorization that moves firms toward less constrained status if they initiate or increase dividends, repurchase shares, or hold very high cash relative to capital expenditures.

When this mixed dependent variable is used, the ordered logit coefficients line up much more closely with KZ. The interpretation is not that KZ is validated. Instead, this shows that KZ is partly produced by putting the same quantitative information on both sides of the regression.

4.4 Evaluating WW and sorting variables

The paper then estimates ordered logits using WW variables and a broad set of common sorting variables.

- In the WW model, cash flow, leverage, and size have signs consistent with the WW index.
- The dividend dummy has the opposite sign from the WW index.
- Industry sales growth and firm sales growth do not consistently predict constraints.
- Bond and commercial-paper rating variables predict constraints in univariate models, but lose explanatory power after controlling for size and age.

Interpretation. Many traditional constraint proxies are useful mainly because they proxy for size and age.

4.5 Constructing the SA index

The paper focuses on firm size and age because they are intuitive, robust, and less endogenous. It estimates ordered logit models with size, size squared, and age. The recommended SA index is:

$$SA = -0.737 \times Size + 0.043 \times Size^2 - 0.040 \times Age. \tag{3}$$

Here $Size$ is the log of inflation-adjusted book assets, capped at the log of \$4.5 billion, and Age is the number of years the firm has been on Compustat with a non-missing stock price, capped at 37 years.

Sign convention. Higher SA values indicate more constraints. Since size and age enter negatively over the relevant range, small and young firms have higher SA values.

4.6 Corroboration using the cash-flow sensitivity of cash

The paper follows Almeida, Campello, and Weisbach by testing whether firms classified as constrained save more cash out of cash flow. The regression outcome is the annual change in cash holdings normalized by assets.

- SA-constrained firms have a positive and significant cash-flow sensitivity of cash.
- SA-unconstrained firms have an insignificant sensitivity.
- KZ-based constrained and unconstrained groups do not cleanly separate this behavior.
- WW has some ability to identify constraints, largely because size is one of its components.

4.7 Investment-cash flow sensitivities

Finally, the paper revisits the Kaplan-Zingales critique. It estimates investment regressions and asks whether investment-cash flow sensitivities increase monotonically with constraint status.

The answer is no. Both the qualitative categories and the SA index fail to produce a monotonic increase. In some specifications, more constrained firms have lower investment-cash flow sensitivities.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies which quantitative firm characteristics line up with hand-coded qualitative disclosures about financial constraints. This is a validation exercise for constraint measures, not a structural model of financing frictions.

5.2 Why qualitative disclosures are useful

Qualitative disclosures provide direct evidence about managers' perceived ability to fund operations and investments. This avoids relying entirely on endogenous financial choices such as cash, leverage, dividends, or payout.

5.3 Limits of qualitative disclosures

The disclosure approach is still imperfect. Managers may use boilerplate language, may underdisclose problems, or may describe debt-contracting events more clearly than non-debt financing frictions. The authors' preferred scheme tries to reduce boilerplate noise by ignoring soft generic warnings.

5.4 The KZ hard-wiring problem

A central identification lesson is that a dependent variable should not embed the same variables used as regressors. If the researcher classifies high-cash firms as unconstrained by construction, the estimated coefficient on cash is partly mechanical.

This matters because the paper's purely qualitative scheme finds the opposite relation for cash: more cash is associated with more constraints, consistent with precautionary cash holdings.

5.5 Endogeneity of leverage and cash flow

Leverage and cash flow do predict qualitative constraints, but the authors are cautious about using them in an index.

- Leverage is a financial choice, not just a constraint outcome.
- Some highly constrained firms may have little debt because they cannot borrow.
- Disclosure-based constraints are often detected through debt events such as covenant violations, so leverage may be mechanically overrepresented.
- Cash flow is also endogenous to operating performance and investment opportunities.

5.6 Why size and age are preferred

Size and age are not perfectly exogenous, but they are much less directly tied to contemporaneous financing choices. They also have a strong intuitive relation to information asymmetry, track record, collateral, public visibility, and access to capital markets.

5.7 How to use the SA index

The SA index is best interpreted as a cross-sectional proxy for financial constraints. It varies slowly because size and age vary slowly. Therefore, when using the SA index in panel regressions, it is prudent to include year fixed effects to absorb aggregate financial conditions.

6 Tables and figures

6.1 Table 1: Frequency of financial constraint categories

Read:

- Table 1 reports the distribution of firm-years across NFC, LNFC, PFC, LFC, and FC categories.
- Under qualitative scheme 1, many firms are placed in the PFC category because of generic warnings.
- Under qualitative scheme 2, the LNFC category rises to 71.59%, while PFC falls to 10.55%.
- The qualitative/quantitative scheme looks much more like Kaplan and Zingales because quantitative variables are used to move firms toward unconstrained categories.

Economic message: The definition of the dependent variable matters. Including quantitative signals inside the classification can make later regressions look supportive of an index for mechanical reasons.

6.2 Table 2: Sample characteristics

Read:

- Table 2 compares all firms, less constrained firms, and more constrained firms.
- More constrained firms have lower cash flow, more leverage, higher Tobin's Q, and are much smaller and younger.
- The median constrained firm has assets of only 14.8 million dollars, compared with 167.1 million dollars for less constrained firms.
- The median age of more constrained firms is 7 years, versus 9 years for less constrained firms.

Economic message: Even before multivariate analysis, size and age are strongly associated with constraint status. Constrained firms look smaller, younger, less profitable, and more levered.

Table 1
Frequency of Financial Constraint Categories

	Constraint assignment procedure			
	Qualitative scheme 1	Qualitative scheme 2	Qual./quant. scheme	KZ sample
Not financially const: NFC	10.28%	10.98%	55.84%	54.50%
Likely not financially const.: LNFC	50.49%	71.59%	31.01%	30.90%
Potentially financially const.: PFC	32.36%	10.55%	6.44%	7.30%
Likely financially const.: LFC	0.32%	0.32%	2.49%	4.80%
Financially const.: FC	6.55%	6.55%	4.22%	2.60%
Correlation with qual. scheme 1	1.00			
Correlation with qual. scheme 2	0.89	1.00		
Correlation with qual./quant. scheme	0.75	0.87	1.00	

This table reports the fraction of all firm-year observations in which an observation is assigned to the indicated financial constraint group. The figures in Columns 1–3 pertain to our random sample of 1,848 Compustat firm years representing 356 firms operating during the 1995–2004 period for observations with non-missing data on the five components of the KZ index. Qualitative scheme 1 uses only qualitative statements made by firms in their filings subsequent to the fiscal year-end regarding the firm's liquidity position and ability to fund investments. The exact algorithm used in coding and categorizing this information is detailed in the text. Qualitative scheme 2 is constructed identically to scheme 1 except that it ignores all soft and generic nonspecific warnings made by firms regarding possible future scenarios under which the firm could experience a liquidity problem. The qualitative/quantitative scheme augments scheme 2 by moving firms upward one category if the firm materially increases dividends, repurchases shares, or has a high (top quartile) level of (cash/capital expenditures) on hand. Additional details concerning the assignment procedures are provided in the text and the Appendix. The figures in Column 4 are taken from table 2 of Kaplan and Zingales (1997) and are based on their sample and algorithm for categorizing constraints. The correlation figures represent simple correlations over the sample between the two constraint assignment procedures in the indicated cell.

Table 1: Frequency of financial constraint categories.

Table 2
Sample Characteristics

Statistic	(1) Mean	(2) Mean	(3) Mean	(4) Median	(5) Median	(6) Median
Cash flow/K	−2.379	−0.915	−9.315	0.243	0.327	−0.907
Cash/K	3.689	3.579	4.208	0.439	0.508	0.199
Dividends/K	0.077	0.064	0.139	0.000	0.000	0.000
Tobin's Q	2.672	2.036	5.686	1.535	1.489	1.809
Debt/total capital	0.338	0.277	0.629	0.275	0.224	0.728
Capital exp./K	0.411	0.415	0.392	0.214	0.229	0.133
Prop., plant, equip. (PPE)	278.370	303.457	159.480	20.664	29.594	2.951
Book assets	782.928	872.877	356.647	124.627	167.089	14.800
Age	13.923	14.716	10.165	9.000	9.000	7.000
Sales growth	0.272	0.247	0.394	0.057	0.070	−0.049
# of qualitative statements	3.37	3.32	3.62	3.00	3.00	3.00
Which observations	All	Less constrained	More constrained	All	Less constrained	More constrained

The figures in each column represent the mean or median of the indicated variable over the indicated set of observations. The figures in Columns 1 and 4 refer to our random sample of 1,848 Compustat firm years representing 356 firms operating during the 1995–2004 period. The figures in Columns 2 and 5 are calculated over the subset of observations in which the firm was classified as less constrained (NFC/LNFC) using qualitative scheme 2 to categorize constraints. The figures in Columns 3 and 6 are calculated over the subset of observations in which the firm was classified as more constrained (PFC/LFC/FC). All variables are constructed from Compustat information. The PPE and book assets statistics are in millions of inflation adjusted year 2004 dollars. All variables that are normalized by K are divided by beginning-of-period PPE. Cash flow is defined to be operating income plus depreciation (Compustat item 18 + item 14). Cash is defined to be cash plus marketable securities (item 1). Dividends are total annual dividend payments (item 21 + item 19). Tobin's Q is defined as (book assets minus book common equity minus deferred taxes plus market equity) / book assets calculated as [item 6 − item 60 − item 74 + (item 25 × item 24)] / item 6. Debt is defined as short-term plus long-term debt (item 9 + item 34). Total capital is defined as debt plus total stockholders' equity (item 9 + item 34 + item 216). If stockholders' equity is negative, we set debt/total capital equal to 1. Capital expenditures are item 128. Age is defined to be the number of years preceding the observation year that the firm has a non-missing stock price on the Compustat file. Sales growth is defined as (sales in year t minus sales in year $t-1$)/sales in year $t-1$. Sales are first inflation adjusted before making this growth calculation. The number of statements row refers to the number of qualitative statements from disclosure filings that were used in assigning the firm to a constraint grouping using qualitative scheme 2, as outlined in the text.

Table 2: Sample characteristics.

6.3 Table 3: Ordered logit models predicting financial constraint status

Read:

- Table 3 evaluates the five KZ components in ordered logit models.
- Cash flow is negative and leverage is positive, consistent with the KZ index.
- The coefficients on Tobin's Q and dividends are less robust.
- Cash is positive and significant in most qualitative specifications, opposite of the KZ index.
- In Column 7, when the dependent variable is partly built from quantitative information, all coefficients line up with KZ.

Economic message: The purely qualitative evidence casts serious doubt on KZ. The model only looks like KZ when the dependent variable is constructed in a way that mechanically favors KZ.

6.4 Table 4: Alternative explanatory variables

Read:

- Table 4 evaluates WW variables, KZ variables one at a time, ACW sorting variables, and age.
- Size and age are consistently negative predictors of constraint status.
- After controlling for size and age, many other variables lose explanatory power.
- Cash flow and leverage remain significant, but the authors hesitate to recommend them because of endogeneity and disclosure-bias concerns.

Economic message: Many popular proxies for constraints are largely proxies for firm maturity. Size and age capture much of the stable cross-sectional variation in constraints.

Table 3
Ordered Logit Models Predicting Financial Constraint Status

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Cash flow/K	−0.008** (0.004)	−0.009** (0.004)	−0.035*** (0.008)	−0.143*** (0.017)	−0.896*** (0.117)	−0.021*** (0.008)	−0.019*** (0.004)
Q	0.037*** (0.009)	0.031*** (0.009)	0.084*** (0.023)	0.099** (0.040)	−0.004 (0.011)	0.037 (0.040)	0.025*** (0.009)
Debt/total capital	1.837*** (0.156)	2.643*** (0.184)	2.758*** (0.195)	2.597*** (0.192)	2.387*** (0.191)	2.777*** (0.228)	2.707*** (0.167)
Dividends/K	0.190 (0.164)	0.224 (0.182)	0.246 (0.334)	−1.171** (0.536)	−1.543 (1.531)	−0.074 (0.292)	−0.513*** (0.195)
Cash/K	0.025*** (0.004)	0.017*** (0.005)	0.019** (0.009)	0.035*** (0.013)	0.284* (0.167)	0.017** (0.007)	−0.022*** (0.006)
Log-likelihood	−1983.929	−1530.061	−1408.514	−1488.404	−1503.759	−1180.758	−1796.799
Pseudo-R^2	0.059	0.096	0.102	0.121	0.112	0.071	0.109
Observations	1848	1848	1752	1848	1848	1562	1848
Specification	Baseline: qualitative scheme 1	Baseline: qualitative scheme 2	Exclude 1% tails	Winsorize at 5%	Normalize by assets	Assets > \$10 million	Baseline: qual./quant scheme

All coefficient estimates are maximum likelihood estimates of an ordered logit model estimated over annual observations. Asymptotic standard errors are reported in parentheses under the coefficients. All estimates are derived from our random sample of 1,848 Compustat firm years representing 356 firms operating during the 1995–2004 period or the indicated subsample. The dependent variable in model 1 (models 2–6) is based on a firm's level of constraints using purely qualitative information following the algorithm described in the text for qualitative scheme 1 (qualitative scheme 2). The dependent variable in model 7 is based on the qualitative/quantitative scheme. In all models, the dependent variable is coded as an integer varying from 1 for the most unconstrained firms (NFC) to 5 for the most constrained (FC) firms. The cash flow, Q, debt, dividends, and cash variables are defined as in the preceding tables and are constructed to be consistent with KZ and Lamont et al. (2001). All fiscal year Compustat variables are matched to corresponding constraints information garnered from filings made immediately following the fiscal year-end. K denotes beginning-of-year property, plant, and equipment. The models in Columns 1, 2, 5, 6, and 7 have all explanatory variables winsorized at the 1% tails. The model in Column 3 excludes all observations with an explanatory variable falling in one of the 1% tails (except for variables that are naturally truncated at a minimum of 0). The model in Column 4 winsorizes all explanatory variables at the 5% tails. The model in Column 5 replaces the K = PPE normalization of the indicated explanatory variables with a K = book assets normalization. The model in Column 6 requires a firm to have book assets exceeding \$10 million (inflation adjusted to 2004).
*Significant at the 10% level.
**Significant at the 5% level.
***Significant at the 1% level.

Table 3: KZ components in ordered logit models.

Table 4
Predicting Financial Constraint Status Using Alternative Explanatory Variables

Specification	(1) WW Vars. together	(2) Individual variables	(3) Indiv. Vars + Size + Age	(4) Parsimonious model
KZ: Cash flow		−0.024*** (0.003)	−0.010*** (0.003)	
KZ: Q		0.071*** (0.008)	0.012 (0.008)	
KZ: Leverage		2.731*** (0.172)	2.858*** (0.181)	
KZ: (Dividends/K)		0.522*** (0.158)	0.156 (0.163)	
KZ: (Cash/K)		0.007 (0.004)	−0.006 (0.005)	
WW: Cash flow	−0.626*** (0.106)	−1.217*** (0.107)	−0.657*** (0.100)	−0.592*** (0.100)
WW: Dividend dummy	0.375*** (0.127)	0.013 (0.113)	0.470*** (0.122)	
WW: Leverage	1.479*** (0.277)	0.804*** (0.242)	1.914*** (0.260)	1.747*** (0.268)
WW: Size	−0.413*** (0.029)	−0.435*** (0.025)	−0.393*** (0.026)	−0.357*** (0.030)
WW: Industry sales growth	0.049 (0.317)	0.016 (0.308)	−0.197 (0.311)	
WW: Firm sales growth	0.014 (0.046)	0.122*** (0.043)	0.042 (0.043)	
ACW: Payout		0.085 (0.134)	0.252* (0.136)	
ACW: Size		−0.0001*** (0.00002)		
ACW: No bond rating		1.791*** (0.202)	0.155 (0.239)	
ACW: No comm. pap. rating		1.911*** (0.226)	−0.085 (0.272)	
Firm age		−0.044*** (0.004)	−0.022*** (0.004)	−0.025*** (0.004)

All estimates are derived from an underlying ordered logit model using firm years as the unit of observation and a dependent variable based on qualitative scheme 2 coded as an integer varying from 1 for NFC firms up to 5 for FC firms. Standard errors are reported in parentheses. All estimates are derived from the sample of 1,848 observations used in the prior tables or the subsample with available information to construct the explanatory variable(s). Each estimate in Column 2 is from a separate ordered logit model in which the indicated variable is the sole explanatory variable. The estimates in Columns 1 and 4 are from single regressions in which all of the indicated variables are included simultaneously. Each estimate in Column 3 is from a separate ordered logit model in which the WW size variable, firm age, and the indicated variable are included together as the only explanatory variables. The size and age coefficients in these Column 3 models are not reported except in the size and age rows in which case we report the coefficients on these two variables when they are included together as the sole two explanatory variables. The KZ variables are defined as in the earlier tables. WW-Cash flow is defined as (operating income plus depreciation)/beginning-of-year book assets. The WW-dividend dummy is a variable indicating positive preferred or common dividends. WW-Leverage is defined as (book value of long-term debt)/current book assets. WW-Size is defined as the log of inflation adjusted (to 2004) assets. Industry sales growth is defined as the most recent annual percentage change in inflation-adjusted three-digit industry sales. Firm sales growth is the firm's most recent annual percentage change in inflation-adjusted sales. ACW-payout is (common dividends + preferred dividends + share repurchases)/operating income. ACW size is the firm's inflation-adjusted book assets. ACW-no bond rating is set equal to 1(0) if a firm has debt in a given year but has no current investment-grade bond rating (has an investment-grade bond rating). This variable is set equal to missing if a firm does not have debt. The ACW commercial paper variable is defined in an analogous manner but indicates the lack/presence of an investment-grade commercial paper rating. Age is defined as the current year minus the first year that the firm has a non-missing stock price on Compustat. All variables are winsorized at the 1% and 99% tails, and leverage and ACW-payout ratios above 1 are set equal to 1. The ACW payout variable is set equal 1 if a firm has negative operating income and positive dividends.
*Significant at the 10% level.
**Significant at the 5% level.
***Significant at the 1% level.

Table 4: Alternative predictors of financial constraint status.

6.5 Table 5: Summary statistics on size, age, and financial constraints

Read:

- Panel A shows that mean constraints decline as firms move from small to large size deciles.
- Mean constraints also decline as firms move from young to old age deciles, though the pattern is less steep than for size.
- Panel B sorts firms into size-age cells. Holding age fixed, larger firms are generally less constrained. Holding size fixed, older firms are generally less constrained.
- The relation is nonlinear: constraints fall sharply for very small or young firms and then flatten for larger and more mature firms.

Economic message: Size and age are not just univariate proxies. They jointly organize the cross-section of financial constraints in a robust and intuitive way.

6.6 Table 6: The role of firm size and age in predicting constraints

Read:

- Table 6 estimates ordered logit models using size, size squared, and age.
- The preferred specification yields the SA index coefficients: -0.737 on size, 0.043 on size squared, and -0.040 on age.
- The size relation is quadratic below the cap and flat for very large firms.
- Age enters linearly after being capped at 37 years.
- The results are stable across subperiods, a 1990 out-of-sample check, leveraged firms only, and year effects.

Economic message: The SA index is not a fragile fitted formula. The size-age relation appears robust across samples and modeling choices.

Table 5
Summary Statistics on Size, Age, and Financial Constraints

Panel A: Mean constraints by sample decile

	Size	Age
Decile 1	3.263	2.309
Decile 2	2.594	2.378
Decile 3	2.324	2.303
Decile 4	2.156	2.435
Decile 5	2.108	2.339
Decile 6	2.023	2.250
Decile 7	1.920	2.286
Decile 8	2.009	2.257
Decile 9	2.000	1.770
Decile 10	1.849	1.851

Panel B: Mean constraints, 5×5 sample sort

	Size: Q1	Size: Q2	Size: Q3	Size: Q4	Size: Q5
Age: Q1	3.045 (111)	2.214 (131)	2.068 (117)	2.073 (82)	2.261 (69)
Age: Q2	3.120 (92)	2.287 (80)	2.200 (65)	2.022 (90)	2.068 (44)
Age: Q3	2.710 (93)	2.369 (84)	2.145 (83)	2.093 (107)	2.040 (50)
Age: Q4	2.852 (122)	2.318 (88)	2.125 (72)	1.782 (87)	1.846 (52)
Age: Q5	2.857 (7)	1.810 (42)	1.841 (88)	1.763 (59)	1.775 (209)

The cells in this table report the mean level of financial constraints for the indicated subsample of observations drawn from our extended sample of 2,124 observations for 382 firms operating during the 1995–2004 period. An observation's constraint level for a given firm year is coded as an integer from 1 to 5 using the qualitative scheme 2 algorithm described in the text, with 1 representing the lowest level of constraints (NFC) and 5 representing the highest level of constraints (FC). Firm size is defined to be the log of assets (inflation adjusted to 2004). Age is defined as the current year minus the first year that the firm has a non-missing stock price on the Compustat file. In panel A, we sort all observations into deciles with decile 1 (decile 10) representing the smallest/youngest (largest/oldest) firms. In panel B, we sort all firms into one of 25 cells based on the size and age quintile that a given observation is a member of. Quintile 1 (Q1) represents the smallest/youngest firms, and quintile 5 (Q5) refers to the larges/oldest firms. In panel B, the number of observations in each quintile is reported in parentheses under the mean financial constraints figure.

Table 5: Size, age, and qualitative financial constraints.

Table 6
The Role of Firm Size and Age in Predicting Constraints

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Size	−0.744*** (0.056)	−0.737*** (0.059)	−1.071*** (0.109)	−0.548*** (0.068)	−1.871** (0.780)	−0.705*** (0.061)	−0.702*** (0.059)	−0.719*** (0.058)
$(\text{Size})^2$	0.042*** (0.007)	0.043*** (0.007)	0.073*** (0.012)	0.025*** (0.009)	0.138* (0.072)	0.040*** (0.007)	0.039*** (0.007)	0.041*** (0.007)
Age	−0.075*** (0.013)	−0.040*** (0.005)	−0.051*** (0.007)	−0.031*** (0.007)	−0.039** (0.019)	−0.041*** (0.005)	−0.040*** (0.005)	−0.040*** (0.005)
$(\text{Age})^2$	0.0010*** (0.0002)							
SEO activity								−10.114 *** (3.110)
Log-likelihood	−1729.494	−1730.412	−846.412	−848.238	−133.058	−1532.780	−1711.870	−1725.088
Pseudo-R^2	0.134	0.133	0.156	0.122	0.092	0.136	0.143	0.136
Observations	2,124	2,124	1,118	1,006	125	1,811	2,124	2124
Sample	All	All	First half 1995–1999	Second half 2000–2004	New 1990 sample	Original, levg. > 0	All, with year effects	All

All coefficient estimates are maximum likelihood estimates from an ordered logit model estimated over annual observations. Asymptotic standard errors are reported in parentheses under the coefficients. The timing convention in this table is the same as in prior tables. The dependent variable in all models is based on a firm's level of constraints using qualitative scheme 2 described in the text. The dependent variable is coded as an integer varying from 1 for the most unconstrained firms (NFC) to 5 for the most constrained (FC) firms. Firm size is defined to be the log of assets (inflation adjusted to 2004). Age is defined as the current year minus the first year that the firm has a non-missing stock price on the Compustat file. In Column 1, size and age are winsorized at the 1% and 99% tails. In Columns 2–8, we winsorize firm size and age at the 1% tails on the low end, and at the $4.5 billion and thirty-seven year points on the high end. SEO activity is the number of SEOs during the calendar year that corresponds to the fiscal year divided by number of public firms on Compustat as of the beginning of the year. The models in Columns 1, 2, 7, and 8 include the entire extended sample from 1995 to 2004 of observations with available data on size, age, and financial constraint status. The model in Column 3 (Column 4) includes the subsample of 1995–1999 (2000–2004) observations. The sample in Column 5 is a new sample of 125 random firms in 1990 with an electronic 10-K or annual report filing available. The sample in Column 6 includes all observations in the entire 1995–2004 sample in which the firm has a nonzero level of leverage. In Column 7, we use the entire sample and add year effects (year estimates not reported) to the model.
*Significant at the 10% level.
**Significant at the 5% level.
***Significant at the 1% level.

Table 6: Size-age ordered logit models and the SA index.

6.7 Table 7: The sensitivity of cash holdings to cash flow

Read:

- Table 7 tests whether constrained firms save cash out of cash flow.
- SA-constrained firms have a cash-flow coefficient of 0.065, significant at the 1% level.
- SA-unconstrained firms have a coefficient near zero and insignificant.
- KZ does not create as clean a split, while WW performs better partly because it includes firm size.

Economic message: An independent test supports the SA index. Firms classified as constrained by SA behave like constrained firms in the cash-saving test.

6.8 Table 8: The sensitivity of investment to cash flow

Read:

- Table 8 revisits investment-cash flow sensitivities.
- Using qualitative categories, the most constrained firms do not have the highest sensitivity.
- The interaction for LFC or FC firms is negative and significant in Column 4.
- Using SA, KZ, or WW indexes also does not produce a monotonic positive relation between constraints and investment-cash flow sensitivity.

Economic message: Investment-cash flow sensitivity is not a reliable constraint measure. This result reinforces the original Kaplan-Zingales critique while also rejecting KZ as an index.

Table 7
The Sensitivity of Cash Holdings to Cash Flow

	(1)	(2)	(3)	(4)	(5)	(6)
Q	0.001 (0.002)	0.002 (0.001)	0.004** (0.001)	0.002 (0.002)	0.002 (0.002)	0.001 (0.001)
Cash flow	0.065*** (0.014)	−0.003 (0.015)	0.022** (0.010)	0.032* (0.019)	0.047*** (0.011)	−0.011 (0.016)
Size	−0.000 (0.003)	−0.001 (0.002)	0.002 (0.001)	0.001 (0.003)	0.001 (0.003)	0.001 (0.002)
Adjusted R^2	0.198	0.064	0.212	0.162	0.188	0.092
# Observations	14,671	14,719	14,601	14,624	14,678	14,701
Which observations	SA index constrained	SA index unconstrained	KZ index constrained	KZ index unconstrained	WW index constrained	WW index unconstrained

The dependent variable in all models is the annual change in a firm's holdings of cash and marketable securities normalized by assets. All models are linear regressions estimated from 1971 to 2004 with firm fixed effects and an error structure that corrects for heteroscedasticity and for within-period error correlation using the White–Huber estimator. The universe of firms included begins with the Compustat universe and imposes certain restrictions outlined in the text and similar to the restrictions imposed by Almeida et al. (2004) (ACW). All model variables are defined as in ACW, and cash flow is normalized by beginning-of-year assets. Q and size (log of inflation adjusted assets) are measured as of the beginning of the observation year, and cash flow is measured contemporaneously with the investment decision. All variables in the regressions of the table are winsorized at the 1% tails. The samples in the odd-numbered (even-numbered) columns are firms with financial constraints index values in the top tercile (bottom tercile) of all observations in the annual cohort measured as of the beginning of the observation year using the indicated financial constraints index.
*Significant at the 10% level.
**Significant at the 5% level.
***Significant at the 1% level.

Table 7: Cash-flow sensitivity of cash by constraint index.

Table 8
The Sensitivity of Investment to Cash Flow

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Q	0.078*** (0.030)	0.023 (0.036)	0.009 (0.017)	0.062*** (0.023)	0.058*** (0.003)	0.058*** (0.003)	0.057*** (0.003)
Cash flow	0.183*** (0.053)	0.223*** (0.070)	0.147* (0.083)	0.185*** (0.044)	0.090*** (0.008)	0.096*** (0.004)	0.110*** (0.008)
Cash flow * (LFC or FC)				−0.129** (0.066)			
Cash flow * medium constraints					0.012 (0.009)	−0.015** (0.007)	−0.008 (0.008)
Cash flow * high constraints					−0.002 (0.009)	−0.034*** (0.011)	−0.027*** (0.009)
Adjusted R^2	0.299	0.223	0.247	0.265	0.204	0.205	0.205
# Observations	979	158	144	1281	32,334	32,122	32,271
Constraints procedure	Qualitative	Qualitative	Qualitative	Qualitative	Size–age index	KZ index	WW index
Which observations	Low constraints	Medium constraints	High constraints	All with qual. data	Compustat 1996–2004	Compustat 1996–2004	Compustat 1996–2004

The dependent variable in all models is a firm's annual spending on capital expenditures normalized by K (beginning-of-year property, plant, and equipment). All coefficients are estimated using OLS, and all models include firm fixed effects and year effects. The cash flow and Q variables are defined as in table 3, and cash flow is normalized by K. Q is measured as of the beginning of the observation year, and cash flow is measured contemporaneously with the investment decision. Observations in Columns 1–4 are for the 1996–2004 period drawn from the original sample of observations with available beginning-of-year qualitative information on financial constraints and non-missing values for the components of the KZ index. In Columns 1–3, we assign each firm to a single constraint category for the entire sample period and estimate a separate regression for the indicated set of firms. The constraint categories (low, medium, and high) are based on the entire set of the firm's annual constraints categorizations using qualitative scheme 2. We assign firms that are in the lowest two annual constraint categories (NFC and LNFC) for all years to the low constraint grouping. We assign firms that are ever in the highest two annual constraint categories (FC and LFC) in any year to the high constraint grouping. All other firms are placed in the medium constraints (Column 2) grouping. The interaction term in Column 4 is cash flow interacted with a dummy variable indicating whether a firm-year observation falls in the LFC or FC financial constraint groups as of the beginning of the observation year. Observations in Columns 5–7 are for the entire Compustat universe of firms over the 1996 to 2004 period excluding financial firms and utilities. In Columns 5–7, we assign each observation to a low, medium, or high constraint grouping depending on the beginning-of-year value of the indicated financial constraints index and the annual tercile breakpoints for the index. All variables in the regressions are winsorized at the 1% tails. In all models, we exclude all observations for which cash flow/$K \leq -1$.
*Significant at the 10% level.
**Significant at the 5% level.
***Significant at the 1% level.

Table 8: Investment-cash flow sensitivities by constraint status.

7 Short summary

- The paper studies how to measure financial constraints.
- It hand-codes qualitative disclosures from annual reports and 10-K filings for a random sample of Compustat firms from 1995 to 2004.
- The preferred dependent variable is a purely qualitative constraint category that avoids using cash, dividends, or repurchases mechanically.
- The KZ index performs poorly against this benchmark.
- Cash holdings enter with the opposite sign from KZ, consistent with constrained firms saving cash for precautionary reasons.
- The KZ index looks more successful only when the dependent variable is partly constructed using quantitative variables.
- Size and age are the strongest and cleanest predictors of constraints.

- The recommended SA index is $-0.737Size + 0.043Size^2 - 0.040Age$, with size and age capped at the specified thresholds.
- SA-constrained firms show a significant cash-flow sensitivity of cash; SA-unconstrained firms do not.
- Investment-cash flow sensitivities do not increase monotonically with constraint status.

8 Conclusion

8.1 Core conclusion

Hadlock and Pierce show that many popular financial-constraint measures are problematic. The KZ index is especially weak when evaluated against a cleaner qualitative benchmark. The paper recommends a simple size-age index as a practical alternative.

8.2 Economic intuition

Financial constraints are most likely to bind for firms with limited track records, high information asymmetry, less collateral, and weaker access to external capital markets. Small and young firms fit this description. Large and mature firms generally have more financing options, more disclosure history, and better capital-market access.

8.3 Incremental contribution revisited

The paper's main contribution is not only proposing SA. It also teaches a measurement lesson: an index can appear validated if the variables used to build it are also embedded in the outcome used to estimate it. This is the key critique of the KZ index.

8.4 Implications

Researchers using financial constraints as a sorting variable should be cautious. The choice of constraint measure can change empirical conclusions. For many corporate finance applications, size and age are transparent, easy to compute, and empirically supported.

8.5 Limitations

The qualitative benchmark depends on managerial disclosure, which can be noisy, incomplete, boilerplate, or biased toward firms with debt-contracting events. Size and age are also imperfect proxies; they capture stable cross-sectional variation better than high-frequency changes in financing frictions.

8.6 Takeaway

The enduring takeaway is simple: move beyond the KZ index. When a clean, practical constraint proxy is needed, the SA index is often more defensible because it relies on robust, intuitive, and relatively exogenous firm characteristics.